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كلية الحاسبات وتقنية المعلومات

Solidus

From Inspection to Insight

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Phase 1: Project Description

1.1. Introduction

For many years, property has always required on-hand evaluations of buildings, which relies on inspectors to be physically at the location and document the conditions of said buildings, where they had to travel and document the condition of the building precisely. With increasing demand on building inspection by home-sellers and buyers caused by their desire for assurance about the state of a building before making a final deal , property inspection services are a crucial part of property domain. [1] And even though this prosses guarantees thoroughness, but unfortunately it can be inefficient ill-suited for the rapid change and increase of demands in the world.

On of the drawbacks of the current system is the lack and shortage of skilled workers, which could eventually lead to the workers being under a lot of pressure from the large quantity of projects that they could be committed to. [2] This can result in lower performance and fewer quality checks and by that more accidents and low-quality buildings will appear. Another downside is scheduling challenges. Using outdated systems or manually documenting the details of all the appointments can result in mistakes like overbooking inspectors and even unnecessary inspection costs and travel time. [2]

To address these issues and shortcomings we need a new approach. Introducing "Solidus", a reliable platform that simplifies inspection requests solves these problems. It is platform that connect with a network of verified and highly qualified staff committed to specific schedules and reasonable prices.

Finally, "Solidus" is not only a booking tool. A vital feature of this platform is its ability to improves customers' experience. The inspectors of it will document their finding via "Solidus" in real-time and the platform will generate a report summarizing what has been submitted by the staff. This new level of transparency and efficiency transforms the inspection process from stress to trust.

1.2. Project Description and Objectives

"Solidus" is an application that provides users whether they are renter, buyer, or banker with the opportunity with the ability to request building inspection. This software deals with a variety of assessments, from structural and plumbing to electrical and safety check ins.

It starts with the user scheduling an appointment they select the date, time, location, contact information and payment method, after clicking on 'Book Inspection Appointment' the app will be directed to the payment page, only then the appointment will be confirmed. Then at 'appointment' tab, the user can view each of appointment and its status (confirmed, canceled, in-prosses). After the inspection appointment is finished, the application will show the users the detailed report written by the staff that shows on the appointment details that contains all of their findings in addition to their suggestions and recommendations.

Objectives:

- **Improves accessibility:** Offers users an easy platform to deal with, that is accessible for users of all diverse backgrounds. To ease their process of applying for home inspection.

- **Enhances efficiency:** 'Solidus' provides a quick and efficient way to book and scheduling appointments online, reducing any holdups.

- **Advocate for sustainability:** Due to its efficiency, it promotes users to embrace safe practices regarding their buildings. Making for better and safer environments.

- **Provides transparency:** The detailed reports given on the application consist of the inspectors' findings and practical suggestions for users to be always informed. Producing stronger trust between the user and service providers.

- **User feedback:** Having users' ratings and feedback leads to constant improvement. It raises users' satisfaction by their opinions being heard and accounted for. And enables the application to make service providers take accountability for their actions creating a much reliable and trustworthy platform.

1.3. Project Team

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1.4. Project Goals

The Solidus platform aims to transform the property inspection process in Saudi Arabia by creating a digital ecosystem that enhances transparency, reliability, and efficiency between clients and certified inspectors. The system will streamline every stage of the inspection cycle from booking and payment to report delivery ensuring that users receive verified, data-driven assessments that reflect the true condition of their properties.

The platform's development focuses on delivering measurable impact within its first operational year through faster report generation, a high rate of verified inspections, and improved communication between all stakeholders. By automating manual processes and maintaining strict verification standards, Solidus aspires to increase inspection credibility and reduce reporting delays, aligning with the digital transformation goals of the Real Estate General Authority and Saudi Vision 2030.

Solidus also emphasizes long-term sustainability and growth. The platform is designed to expand progressively across major cities in the Kingdom, integrating advanced tools such as IoT data analysis, secure digital payments, and performance dashboards for inspectors. This approach ensures a reliable, scalable solution that supports safer real-estate practices and strengthens public trust in property inspection services.

1.5. Scope and Sources of Domain Analysis Information

Solidus is designed to provide comprehensive property inspection solutions for buyers, renters, financial institutions, and inspectors within the real estate sector.

For clients: Solidus offers the ability to create accounts that allow for efficient booking of certified inspectors and access to standardized inspection reports. These reports can be generated in real time following on-site visits or through scheduled

inspections that include detailed photographic evidence, structural evaluations, and health scoring of the property.

For inspectors: Solidus provides a digital workspace that supports the documentation of inspection findings, integration of advanced tools such as thermal imaging and moisture detection, and automated generation of professional reports. This enables inspectors to deliver consistent, transparent, and legally compliant assessments with greater efficiency.

For banks and developers: Solidus ensures access to credible, verifiable inspection data that reduces financial risk and enhances decision-making. Over time, the platform also functions as a repository of property health data, supporting trend analysis, predictive maintenance planning, and policy development for the broader real estate industry.

Sources of Domain Analysis Information

To accurately define the domain and requirements for the Solidus system, several key sources were examined. These sources helped ensure that the app meets user needs and industry standards within the real estate inspection environment.

1. Real Estate Agencies and Inspection Companies:
   Consultations with professional inspection firms and property agencies provided insights into the current challenges in the inspection process such as scheduling inefficiencies, inconsistent report formats, and lack of transparency. Their input guided the design of Solidus's booking, reporting, and verification modules.
2. End Users (Clients, Renters, Buyers, and Banks):
   Feedback from individuals and institutions that frequently require inspections clarified user expectations regarding report accuracy, security, and response time. Their needs shaped key system features like real-time tracking, smart report generation, and secure payment options.
3. Existing Inspection Platforms:
   Analysis of similar systems (e.g., HomeGauge [3], Spectora [4], iAuditor [5]) provided valuable reference points for feature selection, usability design, and integration of technologies like automated PDF reporting and thermal image uploads.

1.6. Features

1.6.1. Included Features

1. Instant Booking System: Clients can book inspections by selecting property type, location, and preferred time through the app.
2. Verified Inspector Profiles: Inspectors are approved and verified, with profiles that display ratings, reviews, certifications, and pricing for transparency.
3. Secure Online Payments: The app supports multiple payment methods including debit cards, credit cards, and digital wallets.
4. Multi-Media Uploads: Inspectors can upload diverse media (photos, videos, audio notes) to provide first-hand property documentation.
5. Smart Report Generation (PDF): Automatically creates professional inspection reports containing photos, notes, and severity level that all of them the inspector provided.
6. Thermal/IoT Data Integration: Supports uploading thermal images, moisture readings, and IoT device outputs directly into inspection reports.
7. Real-Time Status Tracking: Clients can monitor inspection progress (scheduled, in-progress, or report-ready) within the app.
8. Push Notifications and Reminders: Sends alerts for bookings, upcoming inspections, and completed reports.
9. Reports Archive: Provides a secure archive where clients can access all past inspection reports in one place.
10. Ratings and Reviews System: Clients can rate and review inspectors after inspections are completed. These ratings help future users select reliable inspectors.
11. In-App Messaging System: Enables direct communication between clients and inspectors for scheduling updates, clarifications, or follow-ups.
12. Report Feedback & Confirmation: Allows clients to confirm receipt of reports, mark issues as resolved or request re-inspection when needed.

1.6.2. Excluded Features

1. Construction or Repair Services: Solidus does not provide physical repair, renovation, or construction services; it focuses solely on inspections and reporting.
2. Property Sales or Rentals: The platform does not function as a real estate marketplace for buying or renting properties.
3. Independent Valuation or Pricing: Solidus does not provide property market valuations or appraisals; its role is limited to condition assessments.
4. Household IoT Monitoring: The app does not track daily household energy, water, or utility consumption, as its scope is limited to professional inspections.

Phase 2: Business Requirements Specifications

2.1. Domain Analysis

The Solidus system belongs to the real estate inspection domain, which focuses on assessing the condition, safety, and value of properties before purchase, rental, or financing. Traditional inspection processes are often manual, time-consuming, and lack digital traceability [6]. Solidus addresses these issues by introducing a smart, centralized platform that automates the full inspection cycle-from booking to final report generation-while maintaining transparency and efficiency.

The domain involves three primary user groups:

- Clients (Buyers, Renters, Banks): Request and review property inspections through the app.
- Inspectors: Certified professionals who perform the inspection, upload findings, and generate reports.
- Administrators: Manage user accounts, approve inspectors, and monitor platform integrity.

By bridging these groups, Solidus provides a digital ecosystem that supports standardization, faster reporting, and more informed real-estate decisions [3]

2.2. Requirements & Types

2.2.1 Functional Requirements

R1. The system shall allow users to register an account.

- R1.1 The system shall require users to provide their full name, email address, and phone number.
- R1.2 The system shall allow users to set a password and confirm it before account creation.
- R1.3 The system shall send verification code via SMS or email to verify the user's identity.

- R1.4 The system shall allow users to select their role during registration (Client, Inspector, or Administrator).

R2. The system shall allow users to log-in to their accounts.

- R2.1 The system shall authenticate users using their registered email and password.

R3. The system shall allow clients to book inspection appointments.

- R3.1 The system shall allow clients to select the type of property (e.g., apartment, villa, office).
- R3.2 The system shall allow clients to specify location, preferred date, and time.
- R3.3 The system shall display available certified inspectors based on location and schedule.
- R3.4 The system shall changes the status to ”Placed” after successful payment.

R4. The system shall manage secure payments.

- R4.1 The system shall allow users to pay using debit cards, credit cards, or digital wallets.
- R4.2 The system shall confirm successful transactions and generate digital receipts.

R5. The system shall allow inspectors to manage their confirmed appointments.

- R5.1 The system shall display all assigned bookings in the inspector’s dashboard.
- R5.2 The system shall allow inspectors to update the inspection status (Canceled, In-Progress, or Completed) and the system should notify the client.

R6. The system shall allow inspectors to manage their unconfirmed inspections.

- R6.1 The system shall allow inspectors to accept or reject new inspection requests.
- R6.2 The system shall change the status to confirmed if the inspector accept the request and notifies the client.
- R6.1 The system shall notify the client if the inspector rejects the request.

R7. The system shall allow inspectors to upload inspection data and media.

- R7.1 The system shall allow inspectors to upload photos, videos, and notes and suggestions.
- R7.2 The system shall support uploading IoT/thermal data (e.g., temperature, moisture, or energy readings).

- R7.3 The system shall automatically generate a professional inspection report in PDF format containing images, notes, suggestions and measurements and the system should notify the client.

R8. The system shall allow clients to view and download all completed inspection reports.

- R8.1 The system shall display all past reports filtered by date.
- R8.2 The system shall allow downloading reports in PDF format.

R9 The system shall allow administrators to manage accounts and system data.

- R9.1 The administrator shall be able to approve or suspend inspector accounts.
- R9.2 The administrator shall view system statistics and reports.

R10. The system allow users to securely log out of their accounts.

- R10.1 The system shall automatically log out inactive users after a set period.
- R10.2 The system shall clear session data after logout.

R11. The system shall allow clients to rate and review inspectors.

- R11.1 The system shall allow clients to rate inspections on a scale of 1–5 stars.
- R11.2 The system shall allow clients to leave optional written reviews.
- R11.3 The system shall display average ratings on each inspector's public profile.

R12. The system shall include an in-app messaging system.

- R12.1 The system shall allow clients and inspectors to exchange messages regarding booking and inspection details(text, media and location).
- R12.2 The system shall notify users when they receive new messages.

R13. The system shall allow clients to provide feedback or confirm inspection report completion.

- R13.1 The system shall allow clients to mark reports as "Reviewed."
- R13.2 The system shall allow inspectors to view feedback from clients.
- R13.3 The system shall notify inspectors when feedback or confirmation is received.

2.2.2 Non-Functional Requirements

R1. The system shall run on both iOS and Android mobile operating systems.

R2. The system's interface shall be simple, intuitive, and user-friendly for all age groups.

R3. The system shall support two languages: Arabic and English.

R4. The system shall protect all user data and transactions with SSL encryption.

R5. The system shall be capable of handling 10,000 concurrent users without performance degradation.

R6. The system shall respond to user requests within three seconds.

R7. The system shall be available 24/7 except during scheduled maintenance.

R8. The system shall be easy to maintain and update with minimal downtime.

R9. The system shall be accessible in both light and dark themes.

R10. The system shall comply with Saudi data privacy and security regulations [7].

2.3. Techniques for Gathering Data

The information used to determine the requirements of the Solidus system was primarily derived from team discussions, personal observation, and logical analysis of the existing market situation. The team discussed the system objectives from their collective expertise of the inspection process in real estate. In addition to internal brainstorming, the team indicated that property inspection services are still not readily available and in short supply in Saudi Arabia, which vindicated that there was a demand for a better, simple, and dependable digital solution. These conclusions facilitated the identification of the critical features and functionalities required for Solidus.

2.4. Use Case Descriptions

2.4.1. Use Case (1) Description

Table 1. Use Case (1)

Book Appointment	
Brief Description	Allows clients to book a new appointment to inspect their buildings, enter their information and pay for said appointment.
Actors	Client
Pre-conditions	Client has a registered account and a valid payment method.
Basic Flow of Events	1. Client selects "Book Inspection Appointment" option. 2. System displays a form that requests enters building's type, location and the desired date and time. 3. Client fills in the form and submits it. 4. System displays the available inspectors based on the data provided by the client. 5. Client chooses the inspector. 6. System directs the client to the payment page 7. Client fills in their payment information and submits it. 8. System processes the payment securely. 9. System places the request and generates a receipt.
Alternative Flow of Events	If there is no available inspectors, the system displays a message informing the client and redirect them to fill in the form again.
Exceptions	• If the payment does not go through, it notifies the client. • If the inspector rejects the appointment, notifies the client to choose another available inspector.
Post-conditions	The appointment is scheduled successfully.
Special Requirements	Client should be connected to the internet.

2.4.2. Use Case (2) Description

Table 2. Use Case (2).

Process Inspection Requests	
Brief Description	Inspectors can decide if they want to commit to the un-confirmed appointment assigned by the client or not and mange's assigned appointments.
Actors	Inspector
Pre-conditions	• Inspector has a registered account, is verified and does not have an appointment, which overlaps with appointment being offered.
Basic Flow of Events	1. System notifies the inspector of the new appointment request. 2. Inspectors review the details of the appointment and the property. 3. The inspector select "Accept" to approve of the offer. 4. System updates the appointment status and assign the inspector to the appointment.
Alternative Flow of Events	Inspector rejects the request. System notifies the client to choose a new inspector.
Exceptions	If the offer is canceled by the client, the system displays a message to inform the inspector of the unavailability of the offer.
Post-conditions	• Status of the appointment changes to "Confirmed". • The client is notified of the status change.
Special Requirements	The inspector should be connected to the internet.

2.5. Use Case Diagram

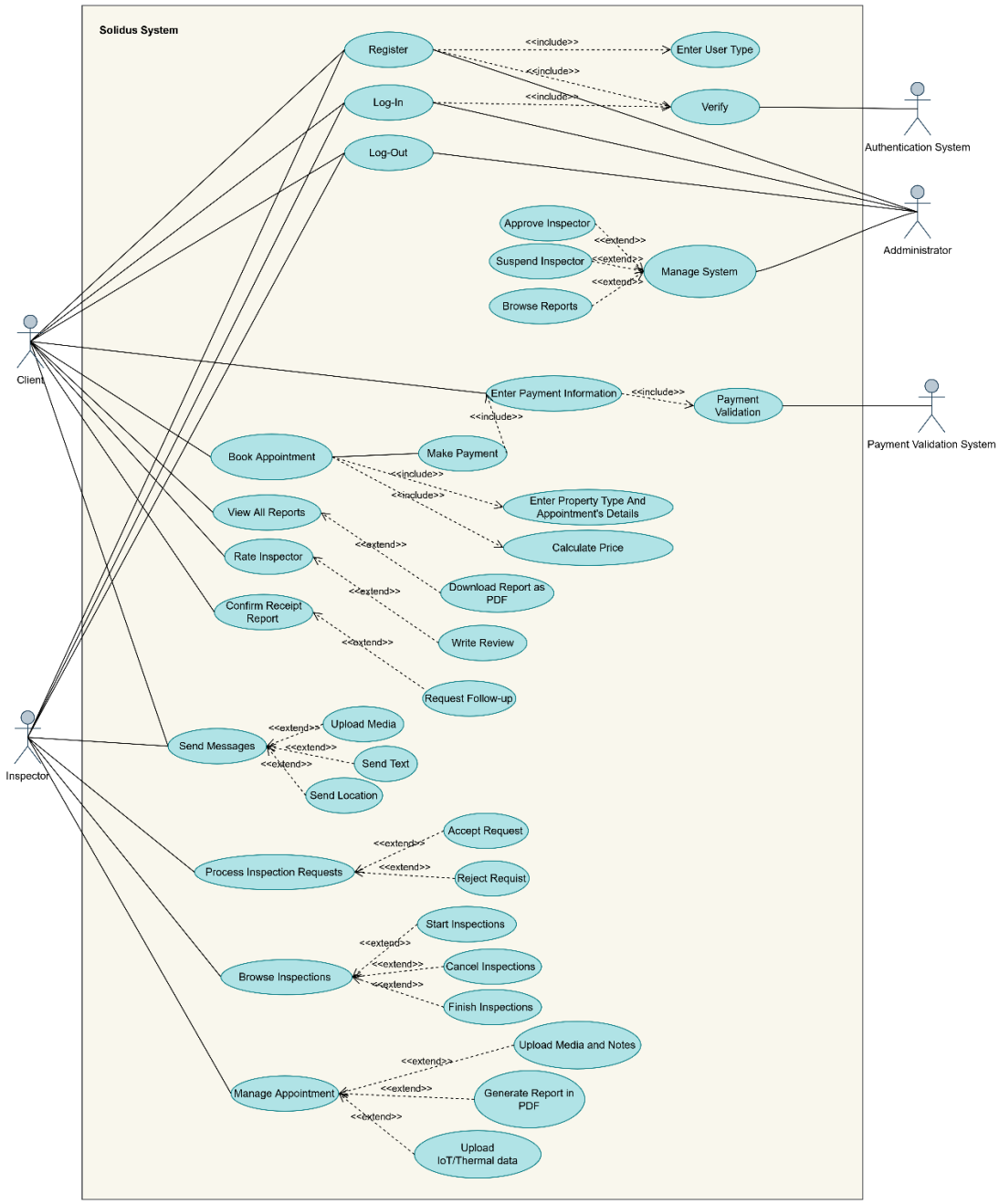


Figure 1. Use Case Diagram

2.6. Difficulties & Risk Analysis

There are a number of potential risks and challenges that could affect the performance and safety of the Solidus system. One of the biggest challenges is that many new features are integrated into one comprehensive system, such as report generation, multimedia uploads, and secure payments. This leads to increased system complexity and greater opportunities for technical errors. Another challenge is the complexity of testing, as there are a multitude of features and user interactions that must be tested in a variety of contexts, such as canceled reservations or failed payment, so that the system performs reliably.

Extended data protection and user privacy are also an important consideration. The system would store sensitive data such as user credentials, property data, and payment information that must be encrypted and stored securely. Weak password security or poorly encrypted data could expose the system to destruction or unauthorized access. The system may also experience performance interruptions if it receives a large increase of users or inspection requests simultaneously, leading to a delayed response time or temporary system outage if not set up adequately.

Additionally, users may experience difficulty in accessing the system in the event of an unstable internet connection or server error that interrupts their booking or reporting process. In order to improve this, the team recommends conducting thorough system testing, enforce password and encryption standards, and employing a well-designed system and secure hosting strategy to guarantee that the proposed Solidus system remains stable, secure and reliable for all users.

Phase 3: Designing and Structuring

3.1. Class Diagram

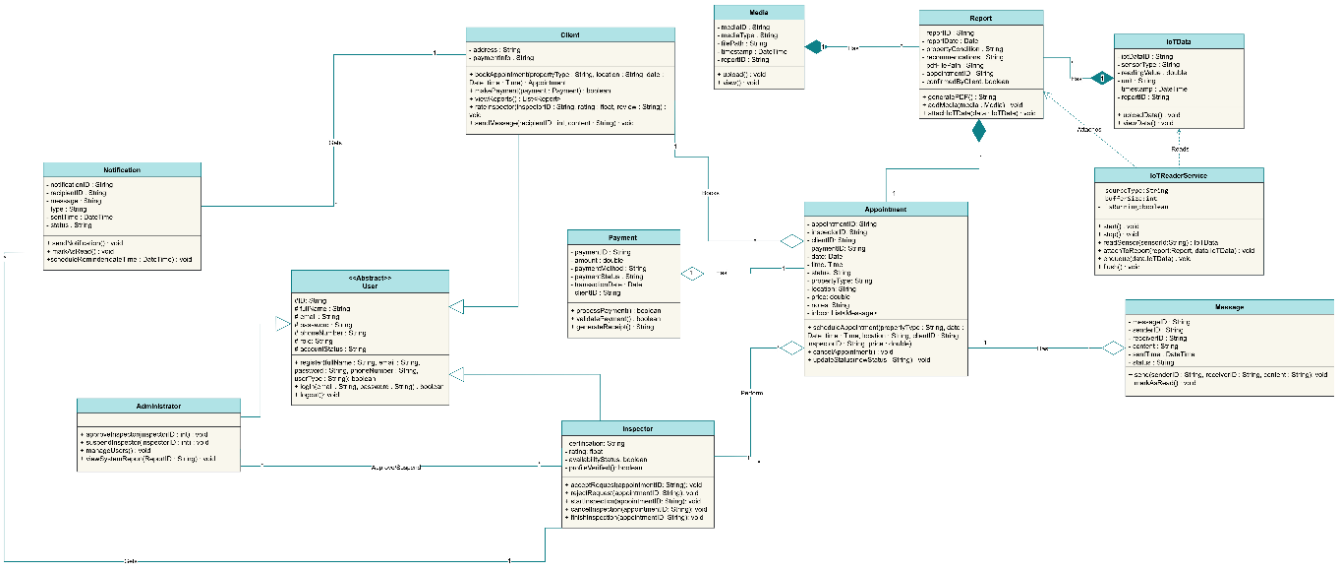


Figure 2. UML class diagram

3.2. Defining Class Associations & Multiplicity

3.2.1. Composition Relationships

- Report class and Appointment class
    - One appointment has one report.
    - One report belongs to one appointment.
- Report class and Media class
    - Each report has many media.
    - Many media belongs to a report.

- Report class and IoTData class
    - Each report may include many IoTData readings
    - Each reading belongs to only one report.

3.2.2. Aggregation Relationships

- Client class and Appointment class
    - Each client can book many appointments.
    - Many appointments can be booked by one client.

- Inspector class and Appointment class
    - Each inspector can perform many appointments.
    - Many appointments can be performed by one inspector.

- Appointment class and Payment class
    - Each appointment has one payment transaction linked to it
    - Each payment transaction is linked to one appointment

- Appointment class and Message class
    - Each appointment has list (many) of messages.
    - Each message belongs to only one appointment.

- Administrator class and Inspector class
    - Many administrators can approve/suspend multiple inspectors.
    - Many inspectors are managed by Many administrators' accounts.

3.2.3. Association Relationships

- Client class and Notification class
    - Each client can receive many notifications.
    - Each notification is addressed to exactly one client.

- Inspector class and Notification class
    - Each inspector can receive many notifications.
    - Each notification is addressed to exactly one inspector.

3.2.4. Generalization Relationships

- Client class and User class
    - Every client is a user.
    - Some users are clients

- Inspector class and User class
    - Every inspector is a user.
    - Some users are inspectors

- Administrator class and User class
    - Every administrator is a user.
    - Some users are administrators

3.2.5. Dependency Relationships

- IoTReaderService class and IoTData class
    - IoTReaderService can read many IoTData.
    - IoTData can be generated by different IoT readers

- IoTReaderService class and Report class
    - Each IoT reader can attach multiple IoT readings to different reports.
    - Each report can receive IoT data from multiple IoT readers.

Phase 4: Modeling, Interaction, and Behavior

4.1. Sequence Diagrams

4.1.1. Sequence Diagram (1)

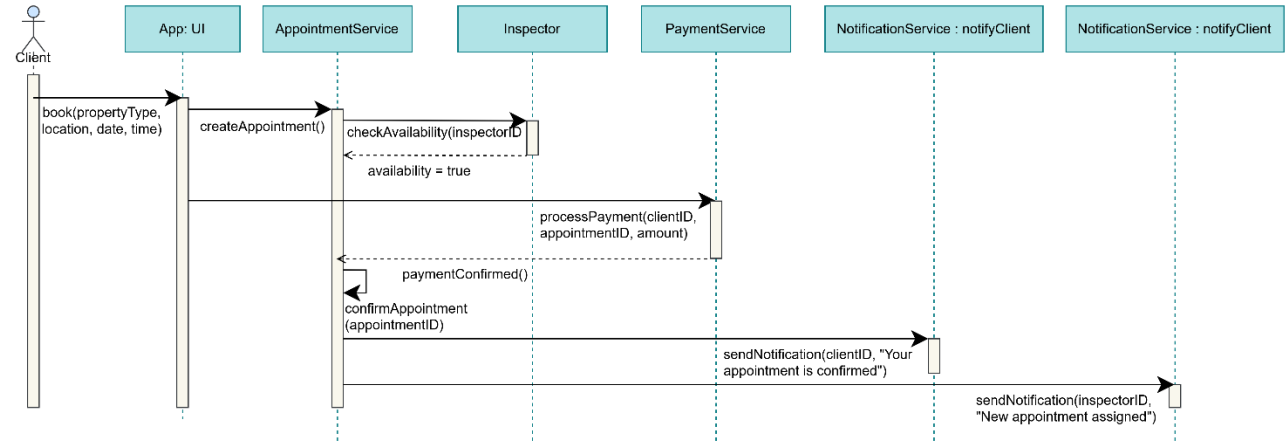


Figure 3. Sequence diagram (1): Client Books an Appointment

4.1.2. Sequence Diagram (2)

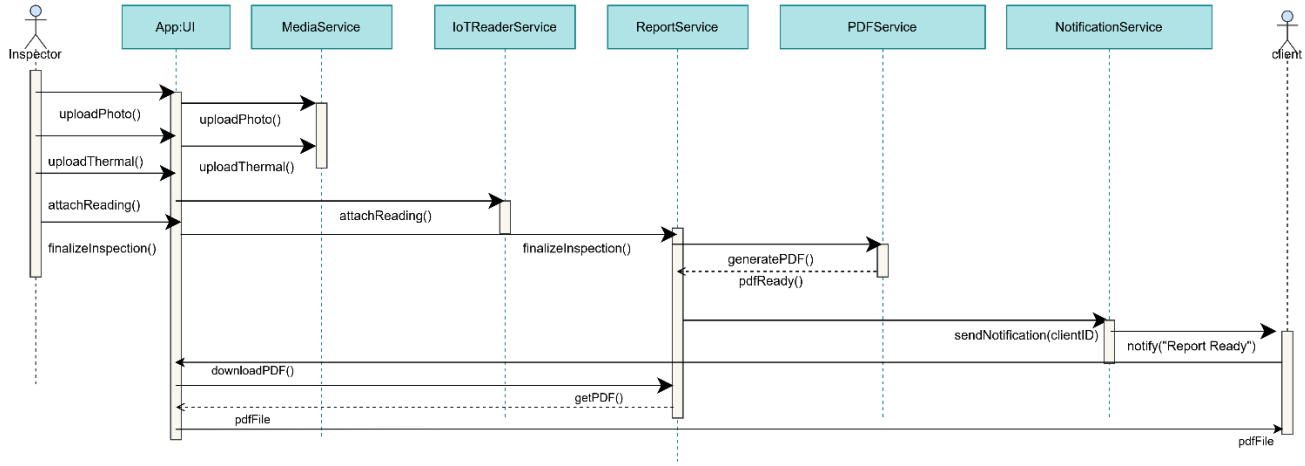
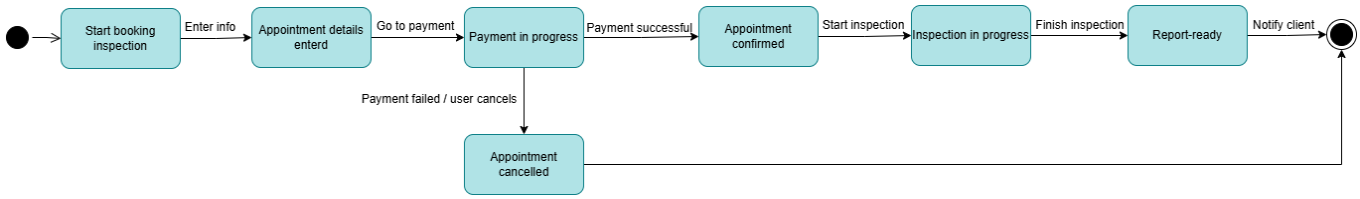


Figure 4 Sequence diagram (2): Inspector Uploads Media

4.2. State Diagrams

4.2.1. State Diagram (1)

- Book Inspection Appointment

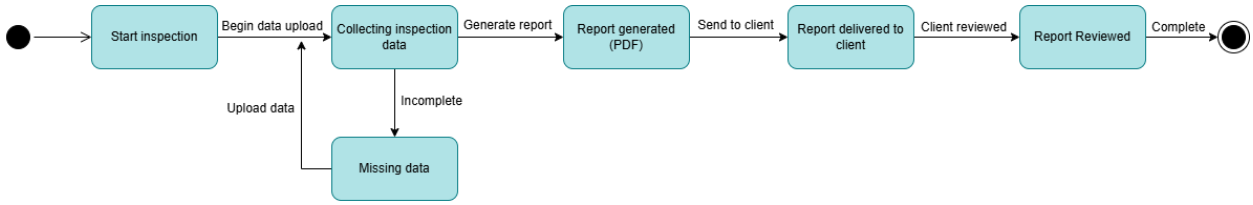


Notes about the first state diagram:

1. The client selects the inspection details such as the date, time, location, and preferred payment method through Solidus.
2. After submitting the booking, the system redirects the user to the payment page to complete the pre-payment for the inspection.
3. If the payment is successful, the appointment status becomes confirmed and later changes to in-progress during the inspection and report-ready once the inspector finishes and submits the report.

4.2.2. State Diagram (2)

- Inspection Report Lifecycle



Notes about the second state diagram:

1. The inspector begins the inspection and starts uploading the required data, including photos, notes, and IoT readings.
2. The system verifies whether all inspection data has been uploaded. If some required information is missing, the inspector is prompted to upload the missing data before proceeding.
3. Once all required data is provided, the system automatically generates the inspection report in PDF format.
4. The generated report is delivered to the client through the Solidus application.
5. The client reviews the report and confirms it, completing the lifecycle of the inspection report.

4.3. Activity Diagrams

4.3.1. Activity Diagram (1)

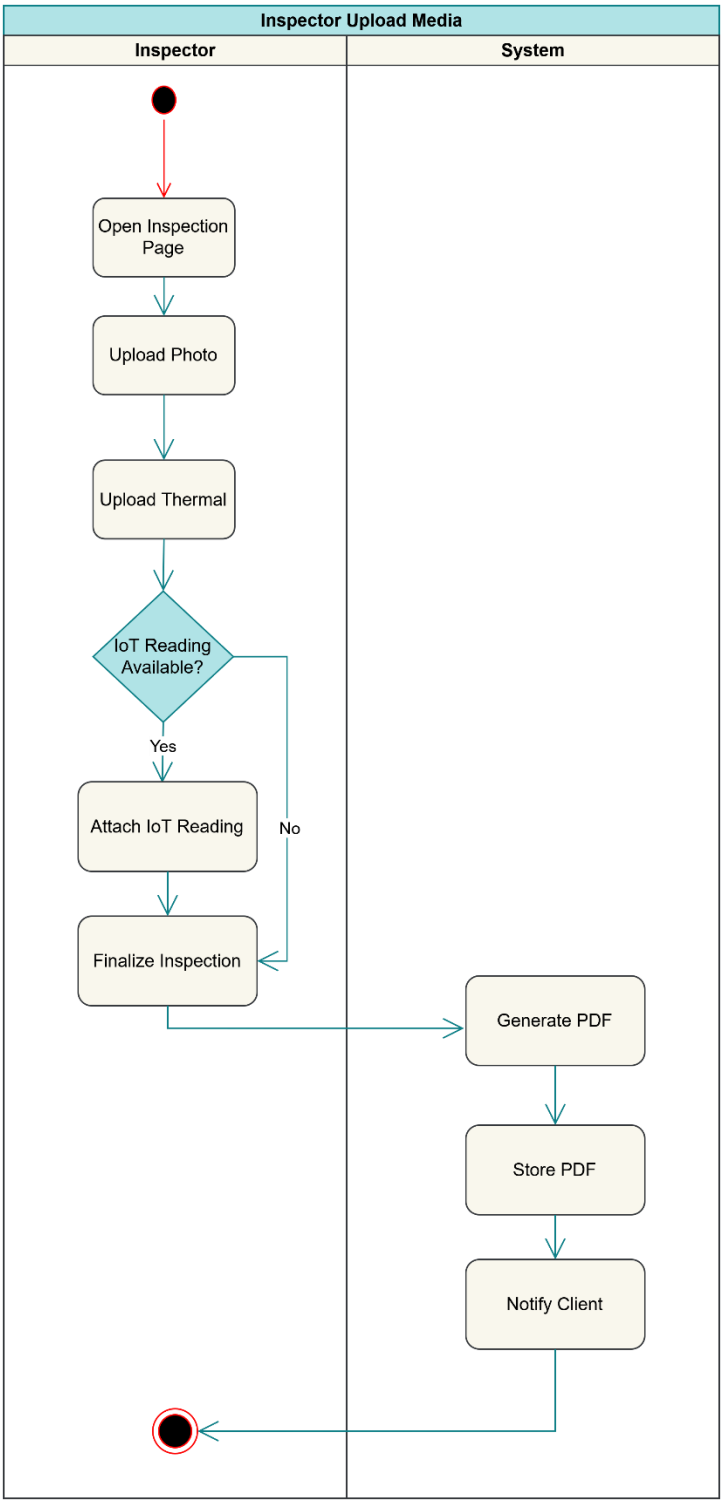


Figure 5 Activity Diagram (1): Inspector Upload Media

4.3.2. Activity Diagram (2)

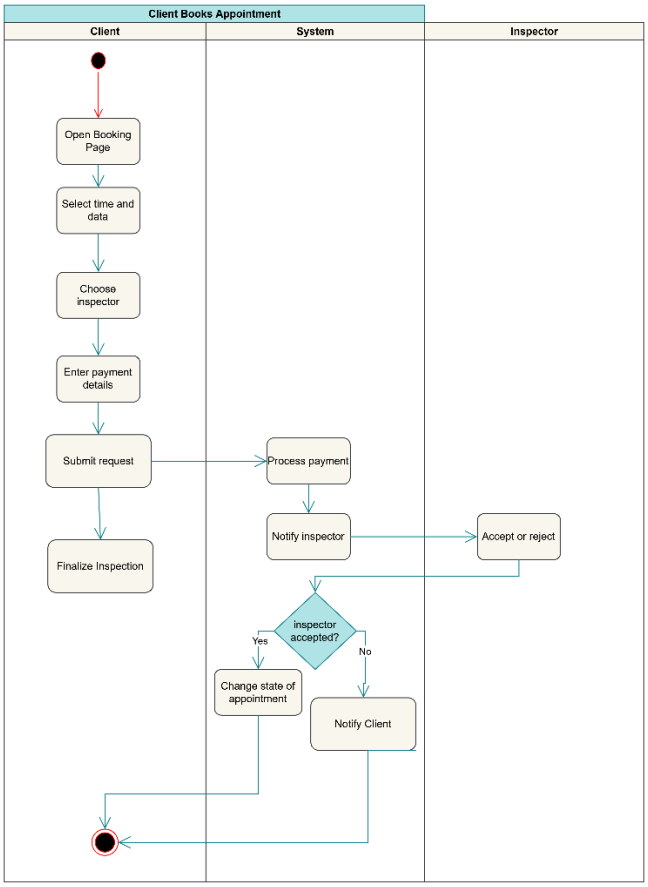


Figure 6 Activity Diagram (2): Client Books Appointment

4.4. System Architecture

The Solidus system follows a multi-tier architecture that separates the presentation layer, application services, and data management to support scalability, maintainability, and secure access control. The architecture is composed of multiple software subsystems that interact together to complete client bookings, process payments, handle inspector tasks, and generate inspection reports.

4.4.1 Components and Subsystems

1. Presentation Layer (Frontend Components)

- **Client Mobile App:**
  Allows clients to register, log in, book inspections, view reports, chat with inspectors, and track inspection status.
- **Inspector Mobile App:**
  Enables inspectors to accept/reject appointments, upload photos and IoT readings, send messages, and finalize reports.
- **Admin Dashboard:**
  Allows administrators to manage inspectors, monitor reports, and review system activities.

2. Application Layer (Backend Services / Subsystems)

- **Authentication & User Management Subsystem:**
  Handles login, registration, verification, and access control for Client, Inspector, and Administrator roles.
- **Appointment Management Subsystem:**
  Manages booking, scheduling, cancellation, and status tracking of inspections.
- **Payment Subsystem:**
  Processes online payments securely and links them to appointments.
- **Messaging Subsystem:**
  Supports real-time communication between clients and inspectors.
- **Media & Report Management Subsystem:**
  Handles media uploads (photos, videos, notes), report creation, PDF generation, and archiving.
- **IoTReaderService Subsystem:**
  Receives IoT/thermal readings from inspection devices and attaches them to reports.
- **Notification Subsystem:**
  Sends booking confirmations, reminders, and report readiness alerts.

3. Data Layer (Storage Components)

- **Relational Database:**
  Stores users, appointments, payments, messages, notifications, reports, and IoT data.
- **Object/File Storage:**
  Stores images, thermal snapshots, videos, and generated PDF reports.

4.4.2 Hardware

Solidus operates mainly as a cloud-hosted software system.
Limited hardware components include:

- **Mobile devices** used by clients and inspectors for capturing media and IoT readings.
- **Cloud server infrastructure** hosting backend services, database, and storage systems. (Physical IoT sensors themselves are not part of the system's hardware scope per project limitations.)

4.4.3 Architectural Pattern

Solidus adopts a **Layered (N-tier) Architecture Pattern**, consisting of:

1. **Presentation Layer** – client and inspector applications.
2. **Application/Service Layer** – business logic and subsystem services.
3. **Data Layer** – database and file storage.

This pattern simplifies maintenance, enforces modularity, and allows independent updates to UI, services, or data storage without affecting the entire system.

4.4.4 System Interaction Overview

- The client application sends booking requests to the Appointment subsystem.
- Inspector availability is checked, and payment is processed through the Payment subsystem.
- During inspection, the IoTReaderService collects sensor readings while MediaService handles photo/video uploads.
- ReportService compiles all data and generates a PDF report stored in object storage.
- NotificationService delivers updates to both client and inspector throughout the process.

4.5. Testing Objectives

The main objective of testing the Solidus system is to ensure that all functional and non-functional requirements are correctly implemented and behave as expected.

Testing focuses on validating system correctness, detecting errors, and ensuring that booking, payments, messaging, report generation, and IoT data processing function reliably under different scenarios.

It also aims to verify security, performance, data accuracy, and user-role access control before full deployment

4.6. Testing Strategy

4.6.1 Levels of Testing

- **Unit Testing:**
  Tests individual components such as AppointmentService, PaymentService, ReportService, IoTReaderService, and NotificationService in isolation.
- **Integration Testing:**
  Ensures correct interaction between subsystems, including the connection between appointments and payments, reports and IoT data, client–inspector messaging, and notification delivery.
- **System Testing:**
  Validates the entire inspection workflow end-to-end, including booking, payment, inspection, report creation, and client confirmation.
- **User Acceptance Testing (UAT):**
  Confirms that the system meets user expectations for usability, accuracy, and reliability.

4.6.2 Testing Techniques

- **Black-box testing:**
  Used to validate system outputs for booking, payment, messaging, and report viewing without analyzing internal code.
- **White-box testing (limited):**
  Applied to critical modules such as payment validation and IoTReaderService data parsing.
- **Boundary value analysis:**
  Applied to input ranges like dates, appointment times, IoT readings, file sizes, and payment amounts.

- **Equivalence partitioning:**
  Used for property types, user roles, and payment methods.
- **Mock testing:**
  Simulates external systems such as payment gateways and IoT device inputs.

4.6.3 Pass/Fail Criteria

- Tests **pass** when:
  - Booking is created and confirmed correctly.
  - Payments are validated and linked to the correct appointment.
  - Reports include all required media and IoT readings.
  - Notifications are delivered to the correct users.
  - Unauthorized access is blocked for all roles.
- Tests **fail** when:
  - Required data is missing or incorrect.
  - Payment cannot be validated.
  - Reports fail to generate or download.
  - Messages or notifications are not delivered.
  - The system violates any functional requirement.

4.6.4 Excluded Features

The following items are excluded from testing during this phase:

- External hardware testing for IoT devices (physical sensors).
- Real payment gateway performance (mocked instead).
- Large-scale stress testing beyond the expected user load.
- Predictive analytics or AI-based features not included in system scope.

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